

Permian Basin Stratigraphy and Tectonic History

Ma	System	Series	Delaware Basin Formations	CBP & NW Shelf Formations	Midland Basin Formations	Tectonic Phase				
252	Permian	Ochoan	Dewey Lake	Dewey Lake	Dewey Lake	final basin fill/termination ↑ Post Orogenic Loading Rapid Subsidence; Onset of HC Generation (~ 50 million yrs. duration) ↓	PERMIAN BASIN PHASE	Active Margin		
			Rustler	Rustler	Rustler					
			Salado	Salado	Salado					
			Castile							
		Guadalupian	Delaware Mtn Group	Lamar	Capitan				Tansill	Tansill
				Bell Canyon					Yates	Yates
				Cherry Canyon					Seven Rivers	Seven Rivers
				Brushy Canyon					Queen	Queen
		Leonardian	Bone Spring	Cutoff Member / 1 st Carb / Avalon	Goat Seep				Grayburg	Grayburg
				1 st Sand					Upper San Andres	San Andres
	2 nd Carb			Lower San Andres						
	2 nd Sand			Lwr San Andres		Holt / Upr Leonard	Holt / Upper Leonard			
	Wolfcampian	Wolfcamp	3 rd Carb / Stray Sd	Yeso / Clearfork	Glorieta	Glorieta				
			3 rd Sand		Upper Clearfork	Upper Clearfork				
					Middle Clearfork	Middle Clearfork				
					Tubb	Tubb				
299	Pennsylvanian	Virgilian	Cisco	Cisco	Cisco	Penn. – Early Permian Foreland Deformation ← <i>Early WFMP Event</i> → ← <i>Strawn Event</i> → ← <i>Atokan Event</i> → ← <i>Morrow Event</i> → (~ 30 million yrs. duration)	PERMIAN BASIN PHASE	Active Margin		
			Missourian	Canyon	Canyon				Canyon	
			Desmoinesian	Strawn	Strawn				Strawn	
			Atokan	Atoka	Atoka				Atoka	
			Morrowan	Morrow	Morrow (NW Shelf Only)				Morrow	
323	Mississippian	Chesterian	Barnett	Barnett	Barnett	↑ Tobosa Basin (Middle Ordovician to Late Mississippian) Moderate Subsidence (~ 150 million yrs. duration) ↓	TOBOSA BASIN PHASE	Passive Margin		
		Meramecian – Osagean	L. Miss Lm	L. Miss Lm	L. Miss Lm					
		Kinderhookian	Woodford	Woodford	Woodford					
359	Devonian	Upper								
		Middle								
		Lower	Devonian (Thirtyone)	Devonian (Thirtyone)	Devonian (Thirtyone)					
419	Silurian	U. Niagaran	Upper Silurian (Wristen)	Upper Silurian (Wristen)	Upper Silurian (Wristen)					
		L. Niagaran								
		Alexandrian	Fusselman	Fusselman	Fusselman					
444	Ordovician	Cincinnatian	Montoya	Montoya	Montoya					
		Mohawkian	Bromide	Bromide	Bromide					
		Chazyan	Tulip Crk.	McKee Sd.	Tulip Crk.				McKee Sd.	
			Mclish	Waddell Sd.	Mclish				Waddell Sd.	
			Oil Creek	Connell Sd.	Oil Creek				Connell Sd.	
			Joins		Joins					
	Canadian	Ellenburger	Ellenburger	Ellenburger						
	Ozarkian									
485	Cambrian	Upper	Bliss	Absent	Wilberns	Great American Carbonate Bank (Late Cambrian – Early Ordovician)				
541	Precambrian		Precambrian Basement	Precambrian Basement	Precambrian Basement	Basement assembly and breakup of Rodinia supercontinent ← <i>Grenville Orogeny (1.3 to 1.0 bya)</i> →				